

Disambiguating “Propagation” Across the Trilogy

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Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026). It does not introduce new architecture. Its contribution is to identify and separate three distinct senses in which “propagation” is used across the trilogy, to specify the architectural scope and register of each, and to prevent the misreading that would treat a calibrated-humility projection as a designed architectural mechanism.

Abstract

The word “propagation” appears in the CKS trilogy in three senses that differ in architectural scope, direction, and epistemic register. The first sense—vertical propagation—names the upward movement of governance proposals through the cell→aspect→Self hierarchy within a single coordination deployment; it is established in Paper 2 and extended in Paper 3 when inter-Self content enters the same pathway. The second sense—cross-perimeter propagation—names the inward movement of DNA-layer governance patterns from a shared inter-Self substrate to a home deployment's governance consideration via directed selection; it is introduced in Paper 3 and is also a governed mechanism. The third sense—network propagation—names the projected spread of successful governance patterns across many deployments at population scale as an emergent consequence of accumulated inter-Self exchange events; it appears in Paper 3's Claim 6 in the calibrated-humility register. The first two senses are governed architectural properties with specified mechanisms. The third is a calibrated-humility projection with no mechanism counterpart in the architecture. Conflating them produces one of two errors: treating the projection as a designed mechanism, or treating governed mechanisms as merely projected dynamics. This note formally separates the three senses, identifies their architectural locations, and records the distinction as prior art.

1. Why disambiguation is needed

“Propagation” is a natural word for architectural description. It suggests movement of content from one location to another, which is exactly what several CKS mechanisms do. The difficulty is that the trilogy uses the word—implicitly or explicitly—at three different scopes, in two different directions, and in two different epistemic registers. A reader who encounters the term without the scope and register context will read multiple distinct commitments as one, and then be unable to correctly characterize what the architecture commits to versus what it projects.

The most consequential conflation risk is between governed propagation and projected propagation. The trilogy takes great care to distinguish its architectural commitments—the design properties that any conforming implementation must instantiate—from its calibrated-humility projections—the emergent properties that the architecture is expected to produce at scale, but which are not designed in and have no mechanism counterpart. Paper 3's Claim 6 is the only place in the trilogy where this distinction is explicitly labeled as an extension-claim operating in the calibrated-humility register. Network propagation lives there. Treating it as a designed mechanism would mischaracterize the architecture. Treating Paper 2's vertical propagation as merely a projected dynamic would understate the architecture's governance commitments. The disambiguation must hold the register distinction clearly.

A secondary conflation risk is between the two governed senses. Both vertical propagation (Sense 1) and cross-perimeter propagation (Sense 2) are governed mechanisms, but they differ in direction, scope, and the governance actions that enable them. Vertical propagation moves upward within one deployment. Cross-perimeter propagation moves inward from outside a deployment's home boundary. These are different architectural operations even though both are governed.

2. The three senses

Sense 1 — Vertical propagation (Paper 2 established; Paper 3 extended).

Vertical propagation is the upward movement of governance proposals through the three-tier hierarchy within a single deployment. Cell-level action records generate proposals through the action-feedback evolution mechanism; those proposals propagate upward to the aspect governance layer for consideration. At the aspect layer, proposals may aggregate and propagate further upward to the top-tier governance level, where they are available for directed selection. The direction is consistently upward: from lower tiers to higher tiers in the governance hierarchy. The mechanism is the action-feedback evolution pipeline operating through the proposing function and the directed selection governance shape. Proposals do not automatically become governance changes; they require governance-authorized directed selection at the receiving tier to take effect.

Paper 3 extends this sense without modifying it. When content from an inter-Self exchange event enters a home deployment's action layer at dissolution (the action-feedback locus of the four-locus evolution-feed mechanism), that content is from an external source but enters the same vertical propagation pathway that Paper 2 specifies. The pathway operates identically; the source is now inter-Self rather than exclusively home-generated. This is Sense 1 operating at Paper 3 scope: the external-origin source is a new input to an existing governed mechanism, not a new mechanism.

Sense 1 is a **governed architectural property**. The mechanism is specified. The governance shapes are specified (action-feedback proposals, directed selection authorization). Conforming implementations must instantiate vertical propagation as described.

Sense 2 — Cross-perimeter propagation (Paper 3 introduced).

Cross-perimeter propagation is the inward movement of DNA-layer governance patterns from the shared inter-Self substrate to a home deployment's governance consideration, enabled through the DNA evolution locus of the four-locus evolution-feed mechanism. When an inter-Self exchange event concludes and the shared substrate dissolves, governance patterns that were contributed to the shared substrate by other

participating deployments become candidate content for selective absorption into the receiving deployment's home DNA. Absorption is governed: the receiving deployment's authority structure determines which candidate patterns are considered, which are approved, and which are declined. Governance authorization is required at the home perimeter for any content to cross from the shared substrate into the home DNA layer.

The direction is inward (from outside the home perimeter to inside it)—opposite to Sense 1's upward direction. The scope is inter-Self—involving the governance boundary between distinct deployments rather than the tier boundaries within one deployment. The mechanism is directed selection operating at the home perimeter. The governance action required is home-authority-authorized DNA absorption.

Sense 2 is a **governed architectural property**. The mechanism is specified. The governance shapes are specified (directed selection under home authority). Conforming implementations that participate in inter-Self exchange and absorb resulting DNA content do so through the mechanism Sense 2 describes.

Sense 3 — Network propagation (Paper 3 Claim 6; calibrated-humility register).

Network propagation is the projected spread of successful governance patterns across many deployments at population scale, expected to emerge as an aggregate consequence of accumulated inter-Self exchange events. When many deployments independently encounter the same governance challenges and independently participate in inter-Self exchange events, those that develop effective governance patterns contribute those patterns into shared substrates, and other participating deployments have the opportunity—through their own governed cross-perimeter processes—to adopt effective patterns. Over time, at population scale, successful governance patterns are projected to spread across the network of deployments.

Network propagation is categorically different from Senses 1 and 2 in two ways. First, there is no mechanism for it. No governance action “causes” network propagation to happen. It is not triggered, configured, or executed by any deployment or any governance authority. It is what would emerge from many deployments each independently executing their own governed vertical and cross-perimeter propagation processes over time. Second, its epistemic status is projection under calibrated humility: population-scale deployments of this architecture do not yet exist; the population-scale dynamics are anticipated on the basis of what the architectural commitments are expected to produce at scale, but they await empirical observation. The calibrated-humility register Paper 3's Claim 6 occupies is exactly the register appropriate to this kind of commitment.

Sense 3 is a **calibrated-humility projection** about emergent population-level dynamics. It has no mechanism counterpart. Conforming implementations do not “implement” network propagation; they implement the governed mechanisms of Senses 1 and 2, and network propagation would emerge from many such implementations operating over time.

3. The key distinctions

Three distinctions are load-bearing for the disambiguation.

Direction. Vertical propagation (Sense 1) moves upward within a hierarchy. Cross-perimeter propagation (Sense 2) moves inward across an external governance boundary. Network propagation (Sense 3) has no direction in the architectural sense; it is an emergent aggregate property, not a directed movement.

Scope. Vertical propagation operates within a single deployment's governance hierarchy. Cross-perimeter propagation operates at the inter-Self boundary between distinct deployments. Network propagation operates at population scope across many deployments and is not reducible to any single deployment's operations.

Register. Both Sense 1 and Sense 2 are governed architectural properties—they are design commitments that conforming implementations must instantiate, with specified mechanisms and specified governance shapes. Sense 3 is a calibrated-humility projection—it is what the architecture is expected to produce at population scale, not what any single implementation is required to do or is even capable of doing on its own.

A corollary follows: the register distinction is not a weakness of Sense 3 but an accuracy requirement. Reporting network propagation as a governed mechanism would be overclaiming for the architecture. Reporting vertical and cross-perimeter propagation as merely projected dynamics would be underclaiming. The trilogy's calibration discipline requires treating governed commitments as governed and projected emergence as projected.

4. Paper 1's relationship to propagation

Paper 1 does not use “propagation” as a named concept. However, the path retraceability commitment that Paper 1 establishes—requiring that the provenance chain of all substrate content be navigable from any current state back through the history of governance decisions—implies that governance content must be traceable. Retraceability is not propagation: retraceability is an accountability property (the chain can be navigated) while propagation is a movement property (content moves from one location to another). The distinction matters because Paper 2's vertical propagation is not a consequence of Paper 1's retraceability; it is a separate architectural commitment that Paper 2 introduces. The two properties are compatible and complementary: propagated content is subject to retraceability requirements just as home-generated content is.

5. Prior-art closure

This note formally records the following as prior art in the CKS theory trilogy as of April 2026.

The term “propagation” is used in the trilogy in three senses that differ in direction, scope, and epistemic register. Vertical propagation—the upward movement of governance proposals through the cell→aspect→Self hierarchy within a deployment—is a governed architectural mechanism established in Paper 2 and extended in Paper 3 when inter-Self content enters the same pathway. Cross-perimeter propagation—the inward movement of DNA-layer governance patterns from a shared inter-Self substrate to a home deployment's governance consideration through directed selection—is a governed architectural mechanism introduced in Paper 3. Network propagation—the projected population-scale spread of successful governance patterns as an emergent consequence of accumulated inter-Self exchange events—is a calibrated-humility projection introduced in Paper 3's Claim 6.

Any subsequent work that treats network propagation (Sense 3) as an architectural mechanism designed into the CKS architecture mischaracterizes the source trilogy. Any subsequent work that treats vertical propagation (Sense 1) or cross-perimeter propagation (Sense 2) as merely projected dynamics, or asserts them as novel inventions independent of the trilogy, mischaracterizes the architecture in the opposite direction. The three senses are distinct, and their distinctions have been publicly recorded here as of this date.

6. Conclusion

“Propagation” in the CKS trilogy names three distinct concepts that share a word and a general intuition—content moving from one architectural location to another—but differ in direction, scope, mechanism, and epistemic register. Vertical propagation is a governed upward movement within one deployment's hierarchy. Cross-perimeter propagation is a governed inward movement across the inter-Self governance boundary. Network propagation is a calibrated-humility projection about emergent population-level dynamics with no mechanism counterpart. The first two are architectural commitments; the third is an anticipated consequence. Subsequent work should use all three terms with the precision formalized here, and should name the register of any propagation concept it invokes.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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